

Two Instruments, One Ecosystem

Open-source bioreactors: what they're *for*, and where OpenEvo goes wrong

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Biopunk Wetware Meeting ▪ `github.com/m9h/biopunk-bioreactor`

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The one-slide thesis

Chi.Bio and OpenEvo look like competitors. They aren't.

- **Chi.Bio is an instrument** — measures a culture richly, in situ, while holding it in a defined state.
- **OpenEvo is an actuator** — pushes a culture through selection, unattended, for weeks.

Both cost ~\$300. Both are open. But a machine built for one purpose is *measurably worse* at the other.

The field serves three purposes, not one contest.

Three purposes

proto – Characterise

Hold a defined state;
measure richly in situ; close
feedback loops.

Champion: **Chi.Bio**

devo – Evolve

Weeks of unattended
selection; produce *archived*,
replicated evidence.

Champion: none yet

bior – Scale out

Many reliable reactors now;
turnkey; supported.

Champions: Pioreactor,
eVOLVER

proto → devo → bior — a lifecycle: prototype, evolve, produce.

Rich-&-few vs. cheap-&-many	non-contact vs. electrodes-OK	steady-state vs.
	run-to-stationary	

Three lineages, one bench

Purpose	Founding lineage	Landmark demo
proto	chemostat as instrument + closed-loop control (Monod; Novick–Szilard; Miliadis-Argeitis '11)	optogenetic feedback holding expression + growth ('16)
devo	chemostat selection → LTEE → PACE (Novick–Szilard '50; Lenski '91; Esvelt '11)	aerobic citrate use in the LTEE ('12)
bior	continuous culture for production → arrays → eVOLVER (Herbert '56; Miller '13; Wong '18)	eVOLVER ePACE: compact Cas9 ('22)

Three traditions, seventy years — now one \$300 device.

OpenEvo: credit where due

A real contribution — these are genuine strengths:

- Peak-to-peak growth readout tied to hardware dilution events (no local-maxima noise).
- 940 nm OD avoids optogenetic & ambient overlap; $R^2 \geq 0.98$ to $OD_{600} = 6$.
- Closed autoclavable vessel — better sterility than Chi.Bio's open tube.
- 4,000-hour remote run; controlled from another continent.
- Assembly manual validated by an undergrad with no electronics experience.

The cheapest buildable three-media selection cyclor in the field.

The problems below are correctable — framing and analysis, not hardware.

OpenEvo: two verifiable errors

Error 1 — the headline figure contradicts itself

Abstract: “**55% faster.**” Results: “4.9 vs 7.6 h, a **36% increase in growth rate.**”

Same data. $7.6 \rightarrow 4.9$ h is a *36% shorter doubling time* = a *55% higher rate* ($1/4.9 \div 1/7.6 = 1.55$). The results sentence mislabels one as the other.

Error 2 — a false claim about a competitor

“Pioreactor. . . only the *software* is open source.”

False: the hardware is openly licensed (CC BY-SA 4.0, per its LICENSE).

(Our own first draft misnamed it CERN-OHL — a contributor corrected it against the primary file.)

OpenEvo: the deeper gap — evidence

$n = 1$

One clone, one population sequenced. No replicates $\Rightarrow$ no convergence $\Rightarrow$ *cannot say which mutation caused the adaptation.*

The LTEE runs **twelve** for this reason.

No archiving

One hand-taken glycerol stock. The LTEE's real innovation was the *freezer*, not the flask — revive an ancestor, race it against a descendant. Without it, fitness is unmeasurable.

OpenEvo cites Barrick's "... Archiving Populations, and Measuring Fitness" — then implements neither.

devo has no champion — yet

OpenEvo is the best *actuator*, not the champion: **devo is defined by evidence, not actuation.**

What the actuator must connect to (only one is new hardware)

- **replicate manifold** (borrow Pioreactor's cluster) → convergence
- **automated chilled archive** — *the one new module* → fossil record
- **sequencing** + **breseq** → correct variant calls
- **fitness loop** (reuses the OD hardware) → selection coefficients
- **selection design** (growth-coupling) → dodge the ALE trap

Champion = mostly connective tissue. Keep the actuator cheap.

The matrix (abridged)

	Chi.Bio	OpenEvo	Pioreactor	eVOLVER	EVE
Purpose	proto	devo	bior	bior	devo
Paper	yes	yes	none	yes	yes
Parallelism	8	1	cluster	16	3+
Rich sensing	spectro.	OD	plugin	config.	OD
Archiving	no	no	no	no	no
Cost	\$300	\$300	\$329	<\$2k/16	\$115
Build	PCB	guided	buy	shop	mod.

Two patterns: **nobody archives** (the field's blind spot) · the only paperless platform is the only pure product.

The proposal: build two, share the parts

Characterisation machine

Non-contact is inviolable.

- + optode DO / pH (read through wall)
- + dispersive spectrometer 340–850 nm
- + sealed gas-controlled lid

Evolution machine

Cheap × many; resist sensing bloat.

- + 8 replicate chambers, one interface
- + **automated chilled archive** (nobody has built this)
- + batch-excursion mode; breseq

Why two? The sensors sort themselves by contact.

Optical (DO/pH/spectrometry) → characterisation. Electrode (conductivity/ERT/viable-cell) → evolution. *Share* the module footprint, sensor designs, firmware, calibration — not the vessel.

Biopunk build I — : an updated Chi.Bio

Chassis: Chi.Bio. *Design law: everything through the wall — no immersed electrodes.*

- **graft** OpenEvo's closed, gas-capable vessel — fixes the open-atmosphere gap
- **add** DO + pH optodes (read through the wall) + Hamamatsu C12880MA dispersive spectrometer
- **port** to Raspberry Pi (all-I²C behind the 0x74 mux — a respin, not a rewrite)

Benefit

The most complete non-contact in-situ characterisation instrument in open hardware — follow metabolic state (DO, pH) and resolve overlapping reporters instead of inferring them.

Pi port → gas vessel → optodes → spectrometer → **8 gas-controlled, closed reactors on one Pi.**

Biopunk build II — **devo**: evolution machine + evidence

Chassis: Pioreactor cluster (or an OpenEvo base for teaching / shoestring).

The champion-maker is the *evidence apparatus* — chassis-agnostic:

- **replicate fleet** (≥ 8 lineages) $\rightarrow$ convergence
- **automated chilled archiver** — **the field's missing freezer; build it first**
- WGS $\rightarrow$ breseq $\rightarrow$ shared causal mutations; revive-ancestor fitness assay
- + OpenEvo 3-media fluidics + closed vessel; conductivity; batch-excursion

Benefit

Fills the empty devo slot — the first open platform that produces *publishable* evolution evidence (causal convergence + measurable fitness), and the substrate for the autopilot.

Prove it at $n \geq 8$ with an archived record = the devo champion $\rightarrow$ then the AI/CRISPR autopilot.

Biopunk build III — **bior**: Pioreactor + add-ons

Chassis: buy Pioreactor. For bior, turnkey reliability *is* the product — mostly configuration, not fabrication.

- **add-on** the shared DO/pH optode module via the plugin path (same footprint — no new board)
- **add-on** eVOLVER Smart Sleeves *only* on the vials that truly need arbitrary per-vial control
- stand on its monthly-maintained software + leader/worker cluster

Benefit

Reliable parallel cultivation now, mostly bought not built — lifted to physiological sensing or per-vial control exactly where a study needs it, nowhere else.

Guardrail: don't over-graft — the value of bior is that you **didn't build it**.

These are teaching instruments too

- **EVE** — eLife paper is literally “... and its educational use”; <\$200; validated in a *real classroom* evolving *E. coli* under chloramphenicol.
- **Pioreactor** — educators portal; one device teaches micro + EE + CS at once.
- **OpenEvo** — the manual *is* a course; anchored a global hackathon.
- **Chi.Bio** — “remove the lid, observe fluorescence by eye.”

But the teaching inherits the research blind spots — archiving, replicates, and carrying capacity are as under-taught as they are under-built.

Where this meets CRISPR

- These tools **evolve** CRISPR — xCas9, base editors, compact Cas9, all PACE / eVOLVER (Hu '18; Thuronyi '19; Huang '22).
- CRISPR runs **inside** these tools — pooled screens in continuous culture; Cas-based growth-coupled selection.
- CRISPR may **close the archiving gap** — in-cell molecular recorders instead of the freezer.

The bior landmark is itself a CRISPR result — the intersection is already here.

The goal: self-driving evolution

OpenEvo *pitched* AI-directed evolution — “AI to supervise the evolution” — but built only the actuator.

The equivalence that makes it buildable

The observation layer an AI needs **is** the evidence apparatus that makes a devo champion.

Apparatus = champion; AI = autopilot. Build the apparatus first, then close the loop.

- **observe:** growth, fitness, genotype (archive → breseq) + in-cell CRISPR recorder
- **act:** media / light / dilution + **CRISPR-targeted mutation** (EvolvR — an IGI tool)
- **learn:** rule-based → Bayesian opt → RL, across the replicate fleet

Prove it: AI-directed vs. fixed schedule, same target, replicate fleet.

`docs/ai-crispr-evolution-project.md`

Take-aways / call to build

- 1 Stop comparing these as one contest — **three purposes**, different winners.
- 2 Evolution has **no champion**: nobody archives or replicates. *That's the opening.*
- 3 The killer feature nobody has shipped: an **automated chilled fraction collector**.
- 4 OpenEvo's errors are real but **fixable** — and we filed them, fairly, with fixes.

Public, CC BY 4.0, corrections welcome — a PR already landed and fixed one of *my* errors.

github.com/m9h/biopunk-bioreactor